

Many Treatments, Some Instruments: Sharp Identification and Specification Testing in General Selection Models

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Scope: Models that impose assumptions on selection, but not outcome

Model

Observables (Y, D, Z) , unobservables $(\mathbf{V}, \varepsilon_Y)$, functions g_D and g_Y :

Treatment Selection Equation: $D = g_D(Z, \mathbf{V})$, (1)

Outcome Equation: $Y = g_Y(D, \mathbf{V}, \varepsilon_Y)$, (2)

Independence Condition: $Z, \mathbf{V}, \varepsilon_Y$ are mutually independent. (3)

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So far, **formally equivalent** to usual potential outcomes IV model, define potential outcomes and treatments:

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Models that impose assumptions on selection, but not outcome

- **Allow:** assumptions on treatment selection equation,
i.e. restrictions on g_D and/or on $\mathbf{V}$
- **Avoid:** assumptions on outcome equation

Selection Models and Response-Types

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Heckman and Pinto (2018): **discrete IV identification problem is finite mixture problem** involving response-type conditional distributions

- **This paper**: **attack mixture problem directly**, develop new results on (partial) identification of components in such mixture models

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For “large” problems where analytic methods may be impractical, I propose computationally feasible algorithm.

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$(Z_M, Z_L) = (0, 0)$: No vouchers

$(Z_M, Z_L) = (0, 1)$: Vouchers for L neighbourhoods

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→ Therefore, $D \in \{H, M, L\}$ and $Z \in \{(0, 0), (0, 1), (1, 1)\}$.

→ Response-type: $T = (D_{(0,0)}, D_{(0,1)}, D_{(1,1)})$.

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Suppose we impose “Unordered Partial Monotonicity” (Mountjoy, 2022):

- Z_M incentivizes $D = M$ and Z_L incentivizes $D = L$
- Changes in Z don't effect choices among **irrelevant** alternatives (e.g., $Z : (0, 0) \rightarrow (0, 1)$ cannot cause $D : H \rightarrow M$)

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Shifting $Z : (1, 1) \rightarrow (0, 1)$ causes two types of moves:

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The following weighted average can be (point) identified

$$\omega_{M \rightarrow L} E \left[Y_L - Y_M \mid D_{(0,1)} = L, D_{(1,1)} = M \right] \\ + \omega_{M \rightarrow H} E \left[Y_H - Y_M \mid D_{(0,1)} = H, D_{(1,1)} = M \right] ,$$

but the two component effects are only partially identified.

Other Examples of Potential Applications

- Multi-dimensional instrument(s) or treatment(s)
 - **Multi-dimensional treatment:** disentangling different dimensions
 - Mediation: want to separate direct, indirect, and total effects
 - More generally: settings with complex incentive schemes, or complicated relationship between Z and D
- Ordered Treatments: e.g., Impact of Incarceration, Schooling etc.
- Nonparametric Discrete Choice Models
 - **MTO, Head Start**, College Choice...
- Other settings with multi-dimensional heterogeneity
 - Survey/RCT Non-Response Bias (e.g., Dutz et al., 2022)
- Exclusion Restriction Violations and/or Sample Selection Problems
 - See, e.g., Kroft, Mourifié, and Vayalinkal (2024)
- Sensitivity analysis for point-identified models
 - In the spirit of Noack (2021)

Related IV Literature

Point Identification: Angrist and Imbens (1995)... Heckman and Vytlacil (1999, 2001, 2007a,b)... Kirkebøen et al. (2016); [Heckman and Pinto \(2018\)](#); Lee and Salanié (2018); Mountjoy (2022)...

Partial Identification: Manski (1989, 1990); Balke and Pearl (1997)... [Beresteanu et al. \(2012\)](#)... Mogstad et al. (2018); Torgovitsky (2019); Kitagawa (2021); Kamat et al. (2023); Marx (2024); [Bai et al. \(2024\)](#)...

- **This paper:** sharp under full IV independence for any selection model; allows multivalued treatments and continuous outcomes; tractable, but not “black box”
- **This paper cannot:** incorporate general assumptions on outcomes, or continuous instruments

Specification Testing : Kitagawa (2015); Mourifié and Wan (2017); Kédagni and Mourifié (2020); Sun (2023)...

- **This paper:** sharp test for general selection models with discrete IV

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Preliminaries: Treatment Selection Model

Definition.

A **treatment selection model** is a 3-tuple $\mathbb{S} = (\mathcal{V}_{\mathbb{S}}, \mathcal{F}_{\mathbb{S}}, \mathcal{G}_{\mathbb{S}})$, where

$\mathcal{V}_{\mathbb{S}}$ is a set ,

$\mathcal{F}_{\mathbb{S}}$ is a subset of $\{F \mid F \text{ is a distribution with } \text{supp}(F) \subseteq \mathcal{V}_{\mathbb{S}}\}$,

and $\mathcal{G}_{\mathbb{S}}$ is a subset of the set of functions $\{\mathcal{Z} \times \mathcal{V}_{\mathbb{S}} \rightarrow \mathcal{D}\}$.

- $\mathcal{V}_{\mathbb{S}}$ represents restrictions on $\text{supp}(V)$,
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$Q \in \mathcal{M}$ **satisfies** $\mathbb{S}$ iff there exists $(Y, D, Z, \mathbf{V}, \epsilon_Y, g_D, g_Y)$ such that:

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- (ii) they satisfy model (1)-(3) and generate Q

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Identification Power of Selection Models

Denote by $\mathcal{Q}_{\mathbb{S}}$ the identified set for parent distributions under $\mathbb{S}$, i.e.

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Define $\mathcal{F}_T(\mathbb{S})$ as set of *a priori* restrictions on distribution of T , i.e.

$$\begin{aligned} \mathcal{F}_T(\mathbb{S}) &:= \left\{ \left(\mathbb{P}_F \left((g_D(z, V) : z \in \mathcal{Z}) = \tau \right) : \tau \in \mathcal{D}^{\mathcal{Z}} \right) \mid g_D \in \mathcal{G}_{\mathbb{S}}, F \in \mathcal{F}_{\mathbb{S}} \right\} \\ &= \{ \pi_Q \mid Q \in \mathcal{M} \text{ and } Q \text{ satisfies } \mathbb{S} \} . \end{aligned}$$

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Theorem 1.

For any treatment selection model $\mathbb{S}$, $\mathcal{Q}_{\mathbb{S}} = \{Q \in \mathcal{Q} \mid \pi_Q \in \mathcal{F}_T(\mathbb{S})\}$.

→ $\mathbb{S}$ has identifying power on features of Q **only** through dist. of T

Parameters of Interest

Today: For simplicity:

- assume **scalar continuous** Y , supported on an interval $\mathcal{Y} \subseteq \mathbb{R}$
- focus on models that simply **eliminate** some response-types
→ i.e. models that restrict

$$\mathcal{T} := \text{supp}(T) = \{\tau_1, \dots, \tau_{N_T}\}$$

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- focus on (weighted averages of) parameters of the form

$$E[Y_d - Y_{d'} \mid \mathbf{T} = \boldsymbol{\tau}],$$

→ examples include **ATE**, **ATT**, **ATUT**, (generalized) **LATE**, **PRTE** etc.

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For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

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→ Observed outcome distributions are mixture distributions:

$$\begin{aligned} \sum_{\tau \in \mathcal{T}} \mathbb{1} \{ \tau(z) = d \} \mathbb{P}(T = \tau) \mathbb{P}(Y_d \leq y \mid T = \tau) \\ = \mathbb{P}(D = d \mid Z = z) \mathbb{P}(Y \leq y \mid D = d, Z = z), \end{aligned}$$

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IV Mixture Model: Basic Idea

For any $d \in \mathcal{D}$ and each $z \in \mathcal{Z}$,

We observe Y for those assigned $Z = z$ and took treatment d

We want to learn about Y_d for those of type $T = \tau$, for some $\tau \in \mathcal{T}$

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→ Allowing Z to vary, get linear system with known coefficients

“Trimming” Bounds of Horowitz and Manski (1995)

Suppose all we observe is the following mixture distribution, with CDF F ,

$$F(y) = \gamma F_1(y) + (1 - \gamma) F_2(y) ,$$

where $\gamma \in (0, 1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

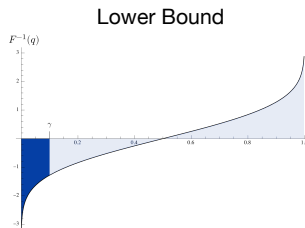
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If $\gamma = 0.1$, sharp lower bound for $E[Y_1]$ is constructed as follows:



(a) Lower bound is area of shaded region

Figure 1: Sharp lower bound for $E[Y_1]$, with $\gamma = 0.1$

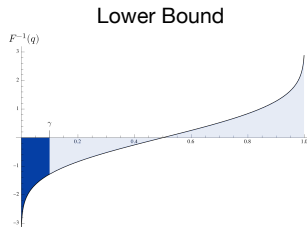
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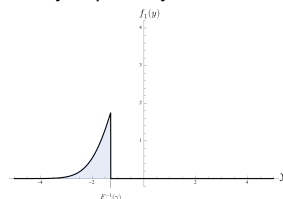
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Density Implied by Lower Bound



(b) Density of F_1

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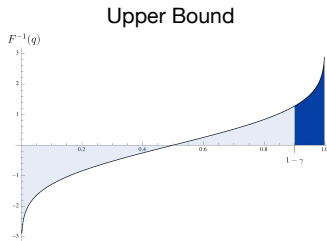
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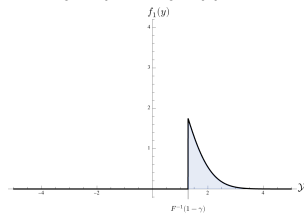
where $\gamma \in (0, 1)$ is known, but F_1 and F_2 are unknown CDFs. Let $Y_1 \sim F_1$.

If $\gamma = 0.1$, sharp **upper** bound for $E[Y_1]$ is constructed as follows:



(a) Upper bound is area of shaded region

Density Implied by Upper Bound



(b) Density of F_1

Figure 1: Sharp upper bound for $E[Y_1]$, with $\gamma = 0.1$

“Trimming” Bounds of Horowitz and Manski (1995)

If instead we have $\gamma = 0.9$, then lower bound is

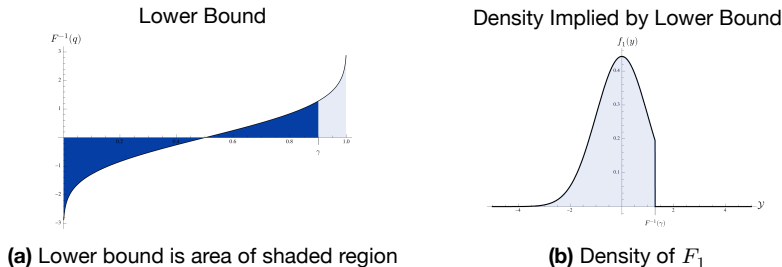


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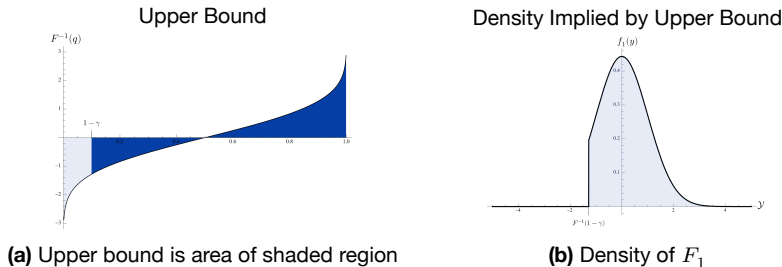


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Horowitz and Manski (1995) Bounds: Takeaway

The observed mixture distribution, written in terms of densities, is

$$f(y) = \gamma f_1(y) + (1 - \gamma) f_2(y) ,$$

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Note that $E[Y_1] := \frac{1}{\gamma} \int_{-\infty}^{\infty} y \gamma f_1(y) d\mu(y)$

→ optimizing the right hand side over all functions $\gamma f_1(\cdot)$ that satisfy the two conditions above results in the Horowitz-Manski bounds

Example 1: Head Start Impact Study

Let D denote preschool choice. Families choose between no preschool (n), competing preschools (c), and Head Start (h) for their child.

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The Head Start Impact Study randomized applicants into two groups:

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→ Therefore, $D \in \{n, c, h\}$ and $Z \in \{0, 1\}$.

→ Response-type: $T = (D_0, D_1)$.

Example 1 (Head Start): Partial Identification

Assume that the following assumption holds for everyone

- either $D_1 = D_0$ or $\mathbb{1}\{D_1 = h\} > \mathbb{1}\{D_0 = h\}$

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Shifting $Z : 0 \rightarrow 1$ causes two types of enrolment changes:

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- Individuals with $T = (n, h)$: $n \rightarrow h$
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The following weighted average can be (point) identified

$$\omega_{n \rightarrow h} E[Y_h - Y_n \mid D_0 = n, D_1 = h] \\ + \omega_{c \rightarrow h} E[Y_h - Y_c \mid D_0 = c, D_1 = h] ,$$

but the two component effects are only partially identified.

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There are two types such that $D_0 \neq h$ and $D_1 = h$:

$$\{(ch), (nh)\}$$

Example 1 (Head Start): Pointwise Restriction

Define

$$\xi_{Y_h|T=(ch)}(y) := \mathbb{P}[T = (ch)] f_{Y_h|T=(ch)}(y) ,$$

and $\xi_{Y_h|T=(nh)}(y) := \mathbb{P}[T = (nh)] f_{Y_h|T=(nh)}(y) .$

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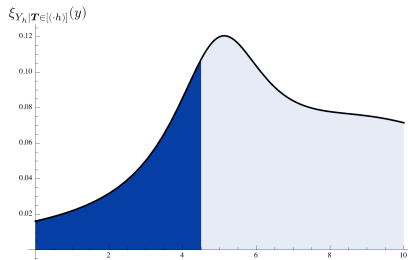
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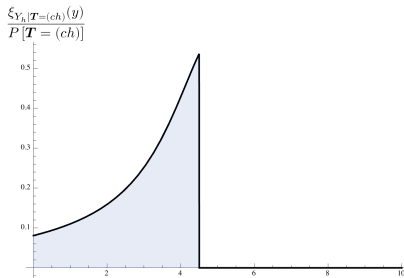
Defining $\xi_{Y_h|T \in [(\cdot h)]} := \xi_{Y_h|T=(ch)}(y) + \xi_{Y_h|T=(nh)}(y)$, we can “trim” the identified function $\xi_{Y_h|T \in [(\cdot h)]}$ to get sharp bounds on $E[Y_h|T = (ch)]$.

Example 1 (Head Start): Sharp Lower Bound

Suppose $\mathbb{P}[T = (ch)] = 0.2$.



(a) Dark blue region has area 0.2.



(b) Density $\frac{\xi_{Y_h|T=(ch)}}{P[T=(ch)]} = f_{Y_h|T=(ch)}$ implied by sharp lower bound

Figure 3: (Simulated) Construction of sharp lower bound for $E[Y_h|T = (ch)]$

Example 2 (MTO): “ M ”-response-type

Recall: $D \in \{H, M, L\}$ and $Z = (Z_M, Z_L) \in \{(0, 0), (0, 1), (1, 1)\}$.

Response-type: $\mathbf{T} = (D_{(0,0)} D_{(0,1)} D_{(1,1)})$.

Consider the effect $\mathbb{E}[Y_H - Y_M \mid \mathbf{T} = (HHM)]$

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There are three types such that $D_{(0,0)} \neq M$, $D_{(0,1)} \neq M$ and $D_{(1,1)} = M$:

$$\{(HHM), (HLM), (LLM)\}$$

Example 2 (MTO): Pointwise Restrictions

In this case, the aggregate density

$$\xi_{Y_M|T \in [(\cdot M)]}(y) := \xi_{Y_M|T=(HHM)}(y) + \xi_{Y_M|T=(HLM)}(y) + \xi_{Y_M|T=(LLM)}(y)$$

is not point-identified, but it can be bounded.

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Define $\bar{f}^*(y)$ to be this bound.

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$$\begin{aligned} & P[D = M|Z = (1, 1)] f_{Y|D=M, Z=(1,1)} - P[D = M|Z = (0, 0)] f_{Y|D=M, Z=(0,0)} \\ &= \xi_{Y_M|T \in [(\cdot M)]} - \xi_{Y_M|T=(MLL)} \\ &\leq \xi_{Y_M|T \in [(\cdot M)]} \end{aligned}$$

Example 2 (MTO): Pointwise Restrictions

In this case, the aggregate density

$$\xi_{Y_M|T \in [(\cdot M)]}(y) := \xi_{Y_M|T=(HHM)}(y) + \xi_{Y_M|T=(HLM)}(y) + \xi_{Y_M|T=(LLM)}(y)$$

is not point-identified, but it can be bounded.

Here,

$$\begin{aligned} & P[D = M|Z = (1, 1)] f_{Y|D=M, Z=(1,1)}(y) \\ &= \xi_{Y_M|T \in [(\cdot M)]}(y) + \xi_{Y_M|T=(MMM)}(y) + \xi_{Y_M|T=(MLM)}(y) . \end{aligned}$$

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Define $\underline{f}^*(y)$ to be the pointwise maximum of 0 and this bound.

Example 2 (MTO): Simple Bounds

So, we have the pointwise bounds $\underline{f}^*(y) \leq \xi_{Y_M|T \in [\cdot, M]}(y) \leq \overline{f}^*(y)$:

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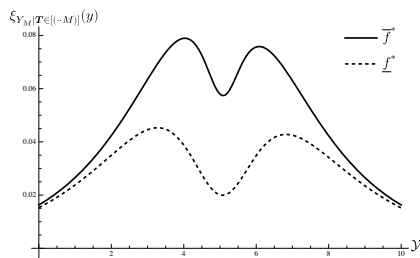
This is because the resulting bound may imply a value of

$$\mathbb{P}[\mathbf{T} \in [(\cdot, M)]] := \mathbb{P}[\mathbf{T} = (HHM)] + \mathbb{P}[\mathbf{T} = (HLM)] + \mathbb{P}[\mathbf{T} = (LLM)]$$

that may not be feasible.

Example 2 (MTO): Non-Sharpness of Simple Bounds

Suppose the sharp upper bound on $\mathbb{P}[T \in [(\cdot \cdot M)]]$ is 0.36, and $\mathbb{P}[T = (HHM)] = 0.28$.



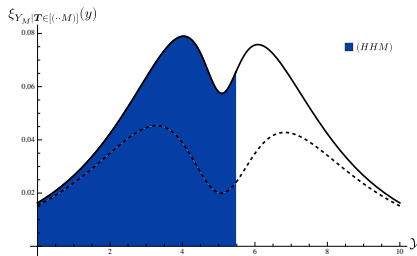
Pointwise bounds on $\xi_{Y_M | T \in [(\cdot \cdot M)]}$

Figure 4: (Simulated) Construction of lower bounds for $E[Y_M | T = (HHM)]$

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Construction of Simple Lower Bound



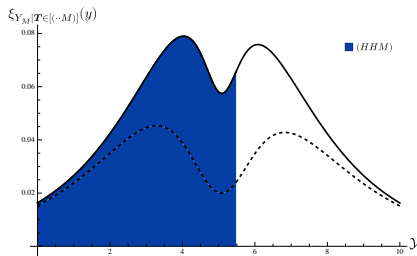
dark blue area = 0.28

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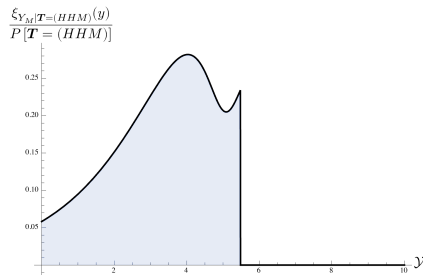
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Construction of Simple Lower Bound



dark blue area = 0.28

Density $f_{Y_M|T=(HHM)}$ Implied by Simple Lower Bound



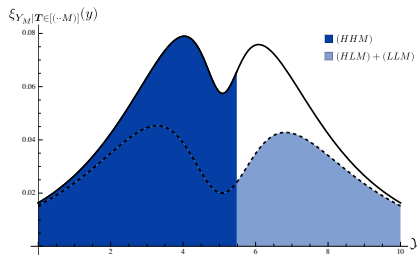
This density implies the simple lower bound of 3.289

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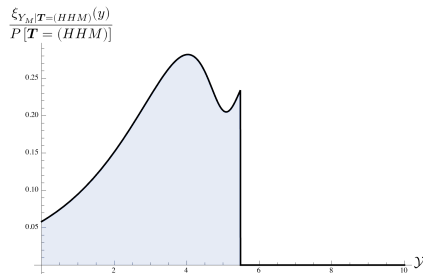
Construction of Simple Lower Bound



dark blue area = 0.28

dark blue area + light blue area = 0.423

Density $f_{Y_M|T=(HHM)}$ Implied by Simple Lower Bound



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Construction of Sharp Lower Bound

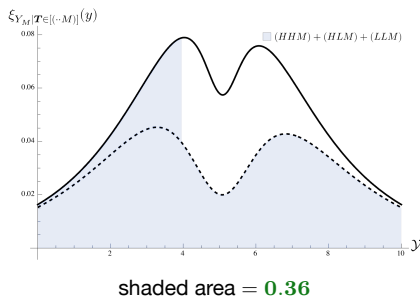
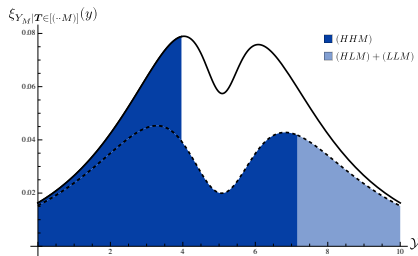


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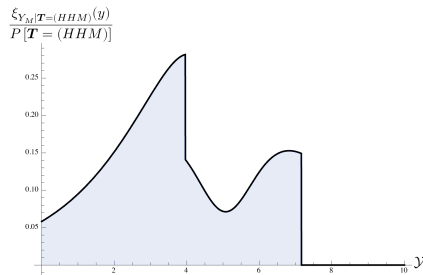
Construction of Sharp Lower Bound



dark blue area = 0.28

dark blue area + light blue area = 0.36

Density $f_{Y_M|T=(HHM)}$ Implied by Sharp Lower Bound



This density implies the sharp lower bound of 3.665

Figure 4: (Simulated) Construction of lower bounds for $E[Y_M|T = (HHM)]$

Example 3: Multi-dimensional Treatment

Consider a setting with a **main treatment**, $M \in \{0, 1\}$, and **supplementary treatment**, $S \in \{0, 1\}$, and define $D = (M, S)$.

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Suppose individuals are

- only offered supplementary incentive if offered main incentive
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→ Therefore, $D \in \{(0, 0), (1, 0), (1, 1)\}$ and $Z \in \{(0, 0), (1, 0), (1, 1)\}$.

→ Response-type: $T = (D_{(0,0)}, D_{(1,0)}, D_{(1,1)})$.

Multi-dimensional Treatment Example

Assume that the following holds for all individuals:

1. $Z_M : 0 \rightarrow 1$ (weakly) increases $\mathbb{1}\{D \in \{(1, 0), (1, 1)\}\}$ and $\mathbb{1}\{S = 1\}$,
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$$\begin{aligned} & \underbrace{\mathbb{E} [Y_{(1,1)} - Y_{(0,0)} \mid \mathbf{T} = ((0, 0), (1, 0), (1, 1))]}_{\text{Total Effect}} \\ &= \underbrace{\mathbb{E} [Y_{(1,1)} - Y_{(1,0)} \mid \mathbf{T} = ((0, 0), (1, 0), (1, 1))]}_{\text{Supplementary Effect}} \\ &+ \underbrace{\mathbb{E} [Y_{(1,0)} - Y_{(0,0)} \mid \mathbf{T} = ((0, 0), (1, 0), (1, 1))]}_{\text{Primary Effect}} \end{aligned}$$

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→ **All components are only partially identified!**

Multi-dim. Treatment example: Singleton “(1, 0)”-response-type

Let's focus on the mean outcome $\mathbb{E} \left[Y_{(1,0)} \mid \boldsymbol{T} = (00, 10, 11) \right]$.

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$$D_{(0,0)} \neq (1, 0), D_{(1,1)} \neq (1, 0) \text{ and } D_{(1,0)} = (1, 0)$$

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Proceeding as before, we have the bounds

$$\begin{aligned} \underline{f}^*(y) &:= \max \left\{ 0, \mathbf{pf}_{(1,0)|Z=(1,0)}(y) - \mathbf{pf}_{(1,0)|Z=(1,1)}(y) - \mathbf{pf}_{(1,0)|Z=(0,0)}(y) \right\} \\ &\leq \xi_{Y_{(1,0)}|\mathbf{T}=(00,10,11)}(y) \leq \\ &\quad \mathbf{pf}_{(1,0)|Z=(1,0)}(y) - \mathbf{pf}_{(1,0)|Z=(1,1)}(y) =: \bar{f}^*(y) , \end{aligned}$$

Multi-dim. Treatment example: Trimming Failure

We have $\underline{f}^*(y) \leq \xi_{Y_{(1,0)}|\mathbf{T}=(00,10,11)}(y) \leq \overline{f}^*(y)$

For simplicity, assume that $\mathbb{P}[\mathbf{T} = (00, 10, 11)]$ is identified, and define

$$\bar{p} := \mathbb{P}[\mathbf{T} = (00, 10, 11)] - \int_{\mathcal{Y}} \underline{f}^*(y) \, d\mu(y) .$$

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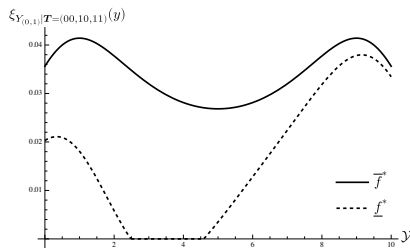
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→ The resulting bounds are valid, but **still not generally sharp**.

Multi-dim. Treatment example: Tighter Bounds due to Pointwise Dependence

Suppose $\mathbb{P}[T = (00, 10, 11)] = 0.25$.



Pointwise bounds on $\xi_{Y_M|T=(00,10,11)}$

Figure 5: Construction of lower bounds for $E[Y_{(1,0)}|T = (00, 10, 11)]$

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Construction of Simple Lower Bound

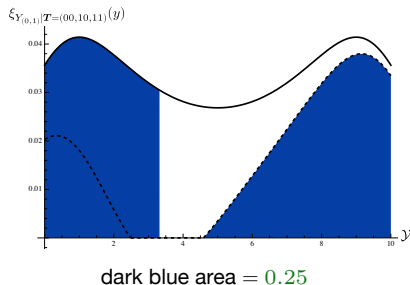
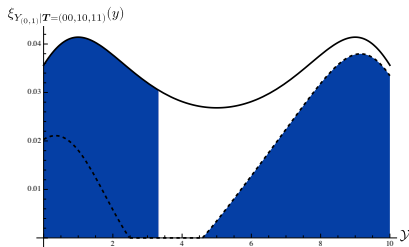


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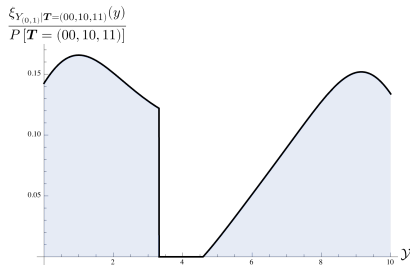
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Construction of Simple Lower Bound



dark blue area = 0.25

Density $f_{Y_{(1,0)}|T=(00,10,11)}$ Implied by
Simple Lower Bound



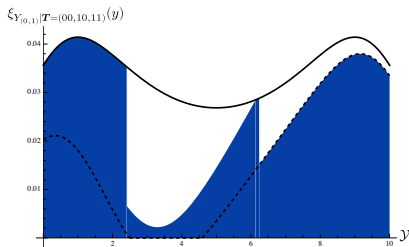
This density implies the simple lower bound
of 4.865

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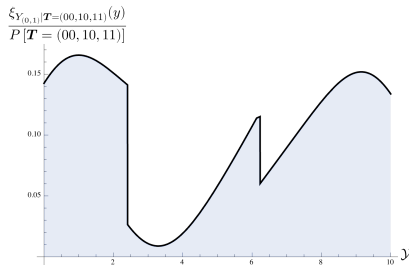
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Construction of Sharp Lower Bound



dark blue area = 0.25

Density $f_{Y_{(1,0)}|T=(00,10,11)}$ Implied by
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This density implies the sharp lower bound
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Figure 5: Construction of lower bounds for $E[Y_{(1,0)}|T = (00, 10, 11)]$

IV Mixture Model: Notation

Enumerate instrument values as $\mathcal{Z} := \{1, \dots, N_Z\}$. For any random variables X_1, X_2 , let $f_{X_1|X_2}$ be the density of $X_1|X_2$.

For $d \in \mathcal{D}$ and $y \in \mathcal{Y}$, define $\mathbf{pf}_d$ as the **observed** vector of weighted densities

$$\mathbf{pf}_d(y) := \begin{bmatrix} P(D = d|Z = 1) f_{Y|D=d,Z=1}(y) \\ \vdots \\ P(D = d|Z = N_Z) f_{Y|D=d,Z=N_Z}(y) \end{bmatrix}$$

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Define $\boldsymbol{\pi}$ as the **unobserved** vector of response-type probabilities,

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IV Mixture Model: Notation

Finally, define the d -**response-matrix** as

$$\mathbf{R}_d := \begin{bmatrix} \mathbb{1} \{ \tau_1 (1) = d \} & \dots & \mathbb{1} \{ \tau_{N_T} (1) = d \} \\ \vdots & \ddots & \vdots \\ \mathbb{1} \{ \tau_0 (N_Z) = d \} & \dots & \mathbb{1} \{ \tau_{N_T} (N_Z) = d \} \end{bmatrix},$$

which is the **known** binary matrix that indicates **when each response-type takes treatment d** (as in Heckman and Pinto, 2018).

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Basic Example ($Z \in \{0, 1\}$, $D \in \{0, 1\}$, no restrictions on selection) : In this case, the response types are always-takers (AT), compliers (C), defiers (D), never-takers (NT) , leading to

$$\mathbf{R}_0 = \begin{matrix} & \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} & \begin{matrix} (Z = 0) \\ (Z = 1) \end{matrix} \end{matrix} \quad \text{and} \quad \mathbf{R}_1 = \begin{matrix} & \begin{matrix} \text{(AT)} & \text{(C)} & \text{(D)} & \text{(NT)} \end{matrix} \\ \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} & \begin{matrix} (Z = 0) \\ (Z = 1) \end{matrix} \end{matrix}$$

IV Mixture Model

Using this notation, we have the following system for $d \in \mathcal{D}$ and $y \in \mathcal{Y}$

$$\mathbf{R}_d \boldsymbol{\xi}_d(y) = \mathbf{p} \mathbf{f}_d(y) \quad \text{and} \quad \boldsymbol{\xi}_d(y) \geq 0 \quad (\star)$$

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Proposition 1 (due to Heckman and Pinto (2018))

For any $\mathbf{c} \in \mathbb{R}^{|\mathcal{T}|}$,

$\mathbf{c} \in \text{row-span}(\mathbf{R}_d) \implies \mathbf{c}^\top \boldsymbol{\xi}_d$ is point-identified ,

$\mathbf{c} \in \text{row-span} \left(\begin{bmatrix} \mathbf{R}_1 \\ \vdots \\ \mathbf{R}_{N_D} \end{bmatrix} \right) \implies \mathbf{c}^\top \boldsymbol{\pi}$ is point-identified .

→ Equivalent to Theorem T-2 in Heckman and Pinto (2018), where it is stated in terms of the Moore-Penrose generalized inverse of $\mathbf{R}_d$

Informal Outline of Approach

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Step 3: Define parameter of interest (e.g., ATE)

- Possible values for this parameter are those consistent with restrictions from Steps 1 and 2

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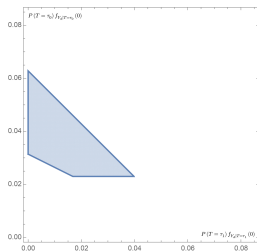
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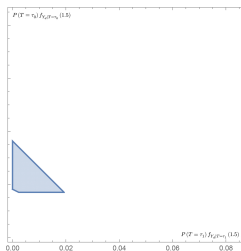
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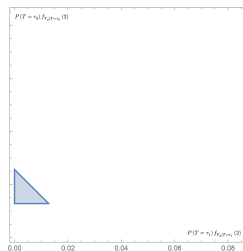
→ By $(\star)$, we require that $\xi_d(y) \in \mathcal{B}_d(\mathbf{pf}_d(y))$ for each $y \in \mathcal{Y}$.



(a) $\tilde{\mathcal{B}}_d(\mathbf{pf}_d(0))$



(b) $\tilde{\mathcal{B}}_d(\mathbf{pf}_d(1.5))$



(c) $\tilde{\mathcal{B}}_d(\mathbf{pf}_d(2))$

Figure 6: Hypothetical $\tilde{\mathcal{B}}_d(\mathbf{pf}_d(\cdot))$ in a case with 2 “degrees of freedom”

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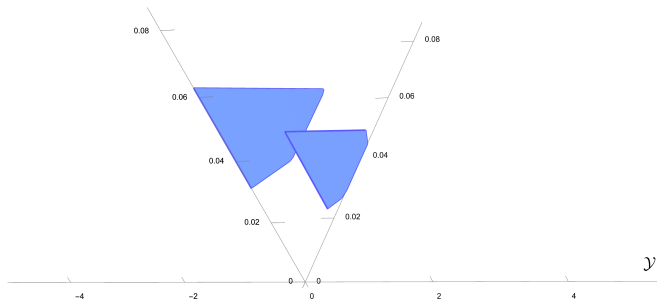


Figure 7: Hypothetical graph of the correspondence $y \mapsto \tilde{\mathcal{B}}_d(\mathbf{pf}_d(y))$

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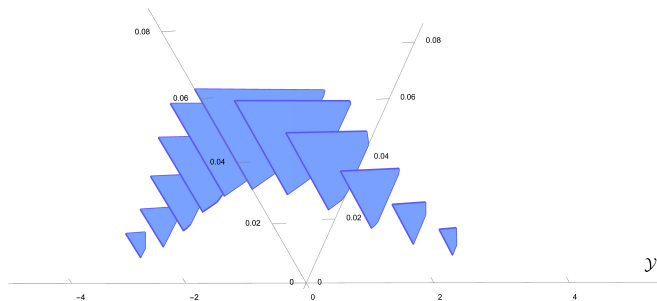


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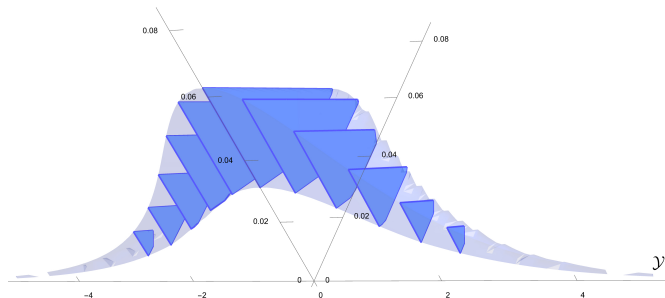


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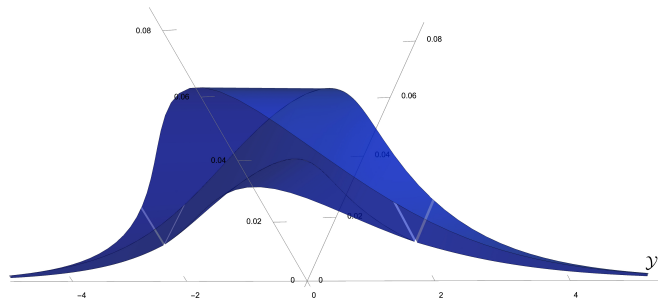


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→ If $\mathbf{x}$ is a vector-valued function such that $\mathbf{x}(y) \in \mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))$ for all $y \in \mathcal{Y}$, then $\mathbf{x}$ is a “candidate” for $\xi_d(y)$.

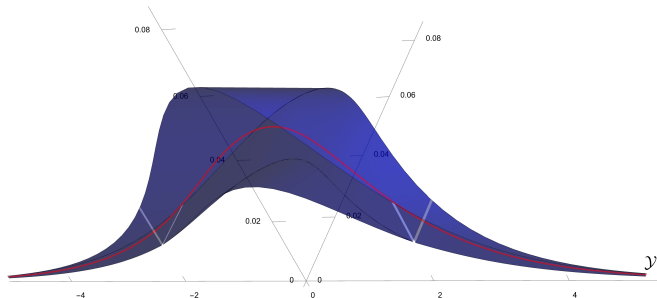


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Candidates for π

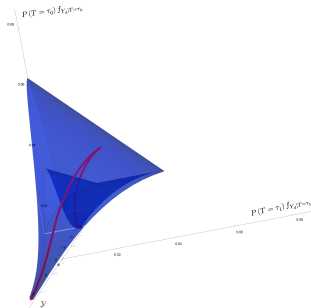
If $\mathbf{x}$ is a candidate for $\xi_d(y)$, then $\int_{\mathcal{Y}} \mathbf{x}(y) \, d\mu(y)$ is a candidate for π , the vector of response-type probabilities. Precisely,

$$\mathbf{x} = \xi_d \implies \int_{\mathcal{Y}} \mathbf{x}(y) \, d\mu(y) = \pi$$

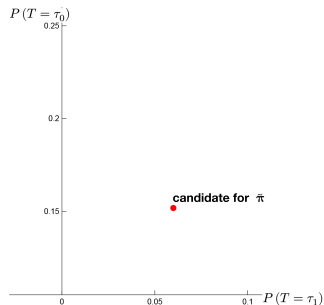
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Define $\Pi_{\mathcal{T}|d}$ as the set of such candidates for π :

$$\begin{aligned} \Pi_{\mathcal{T}|d} &:= \left\{ \int_{\mathcal{Y}} \mathbf{x}(y) \, d\mu(y) \mid \begin{array}{l} \text{measurable } \mathbf{x} : \mathcal{Y} \rightarrow \mathbb{R}_{\geq 0}^{\mathcal{T}} \\ \mathbf{R}_d \mathbf{x}(y) = \mathbf{p} \mathbf{f}_d(y), \, \forall y \in \mathcal{Y} \end{array} \right\} \\ &= \left\{ \int_{\mathcal{Y}} \mathbf{x}(y) \, d\mu(y) \mid \begin{array}{l} \text{measurable } \mathbf{x} : \mathcal{Y} \rightarrow \mathbb{R}^{\mathcal{T}} \\ \mathbf{x}(y) \in \mathcal{B}_d(\mathbf{p} \mathbf{f}_d(y)), \, \forall y \in \mathcal{Y} \end{array} \right\}. \end{aligned}$$

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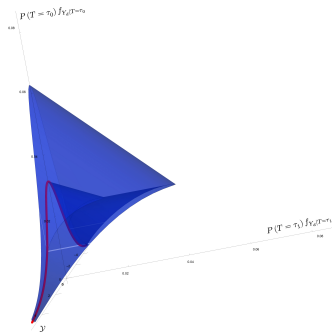
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$\rightarrow \Pi_{\mathcal{T}|d}$ is a convex polytope, and its boundary can be obtained by integrating along the boundary of $\mathcal{B}_d(\mathbf{p} \mathbf{f}_d(y))$

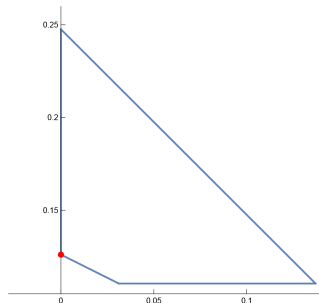
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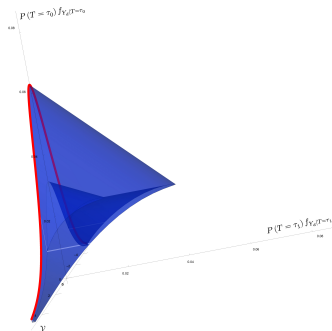


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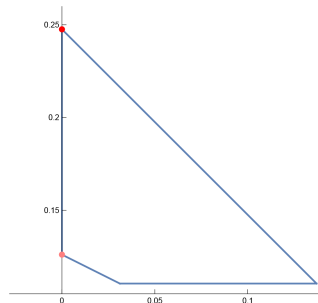
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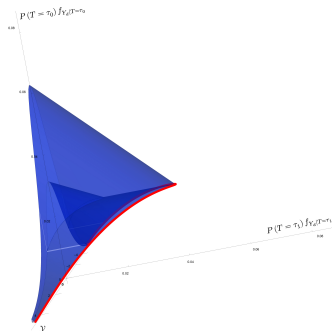


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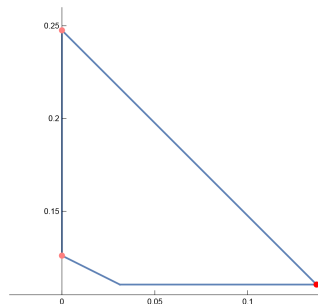
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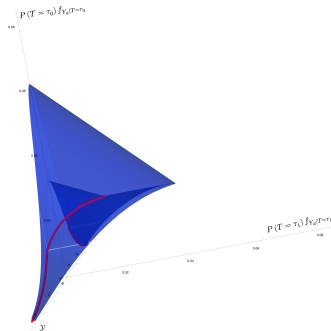


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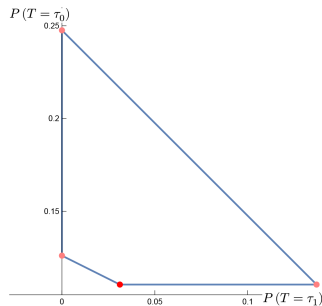
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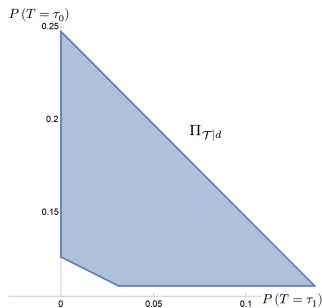


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 - Close connection to *Aumann expectation* from theory of random sets (Artstein and Vitale, 1975): both are adaptations of Aumann (1965)'s integral for set-valued measures

Candidates for π : Support Function

Every non-empty closed convex set $C \subseteq \mathbb{R}^n$ is uniquely determined by its **support function** $h_C : \mathbb{R}^n \rightarrow \mathbb{R}$, defined as

$$h_C(\mathbf{c}) := \sup_{\mathbf{x} \in C} \mathbf{c}^\top \mathbf{x} .$$

Define the shorthand $\mathbf{h}_d(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) := h_{\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y))}(\mathbf{c})$.

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Lemma 1

Suppose $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$ is such that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$. Then $\Pi_{\mathcal{T}|d}$ is a non-empty convex set such that, for all $\mathbf{c} \in \mathbb{R}^{\mathcal{T}}$

$$h_{\Pi_{\mathcal{T}|d}}(\mathbf{c}) = \int_{\mathcal{Y}} h_d(\mathbf{c}; \mathbf{p}\mathbf{f}_d(y)) \, d\mu(y) .$$

Further, $\Pi_{\mathcal{T}|d}$ is a convex polytope.

Candidates for π : Support Function

Corollary 1

Suppose $\mathcal{T} \subseteq \mathcal{D}^{\mathcal{Z}}$ is such that $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset$ for all $y \in \mathcal{Y}$.

Given matrix $\mathbf{A} = \begin{bmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_n^\top \end{bmatrix} \in \mathbb{R}^{n \times \mathcal{T}}$, such that, for some $\mathbf{b} : \mathcal{Y} \rightarrow \mathbb{R}^n$,

$$\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) = \{\mathbf{x} \in \mathbb{R}^{\mathcal{T}} \mid \mathbf{A}\mathbf{x} \leq \mathbf{b}(y)\} \quad \forall y \in \mathcal{Y},$$

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Identified Set for Response-Type Probabilities

Given selection model $\mathbb{S}$, denote by $\mathcal{T}(\mathbb{S})$ the set of response-types not *a priori* eliminated by $\mathbb{S}$, i.e

$$\mathcal{T}(\mathbb{S}) := \left\{ \tau \in \mathcal{D}^{\mathcal{Z}} \mid \exists x \in \mathcal{F}_T(\mathbb{S}), x_{(\tau)} > 0 \right\} .$$

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Theorem 2 (Sharp Testable Implications of Selection Models)

For any treatment selection model $\mathbb{S}$,

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Given selection model $\mathbb{S}$, denote by $\mathcal{T}(\mathbb{S})$ the set of response-types *not a priori* eliminated by $\mathbb{S}$, i.e

$$\mathcal{T}(\mathbb{S}) := \left\{ \tau \in \mathcal{D}^Z \mid \exists x \in \mathcal{F}_T(\mathbb{S}), x_{(\tau)} > 0 \right\} .$$

Denote by $\Pi_{\mathbb{S}}$ the *identified set for π* under $\mathbb{S}$, i.e. $\Pi_{\mathbb{S}} := \{ \pi_Q \mid Q \in \mathcal{Q}_{\mathbb{S}} \}$

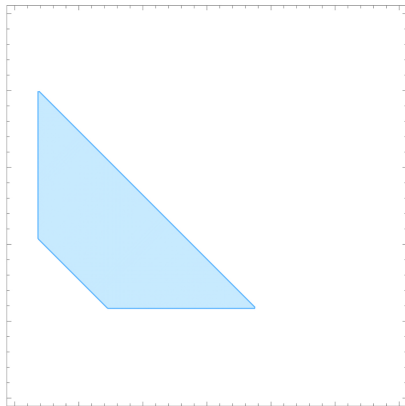
Theorem 2 (Sharp Testable Implications of Selection Models)

For any treatment selection model $\mathbb{S}$,

$$\Pi_{\mathbb{S}} = \left(\bigcap_{d \in \mathcal{D}} \Pi_{\mathcal{T}(\mathbb{S})|d} \right) \cap \mathcal{F}_T(\mathbb{S}) .$$

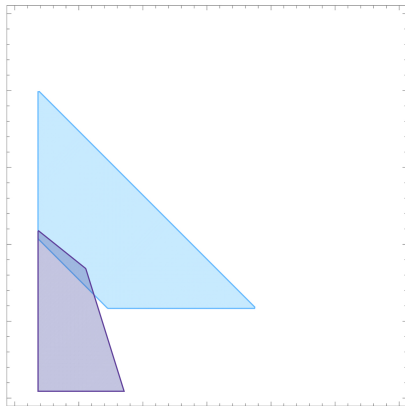
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(Stylized) Example with 3 Treatments



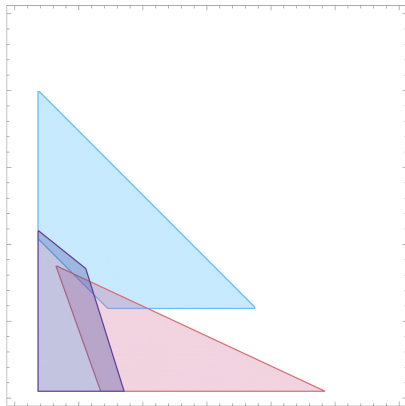
(a) Hypothetical $\Pi_{\mathcal{T}|1}$

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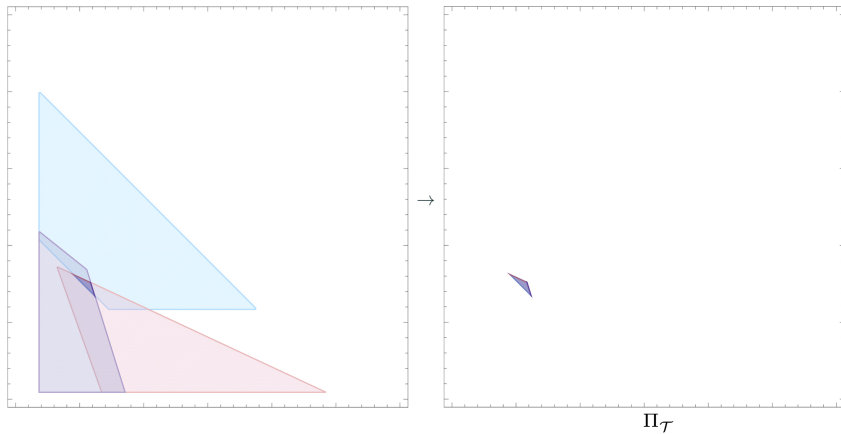
(a) Hypothetical $\Pi_{\mathcal{T}|1}, \Pi_{\mathcal{T}|2}$

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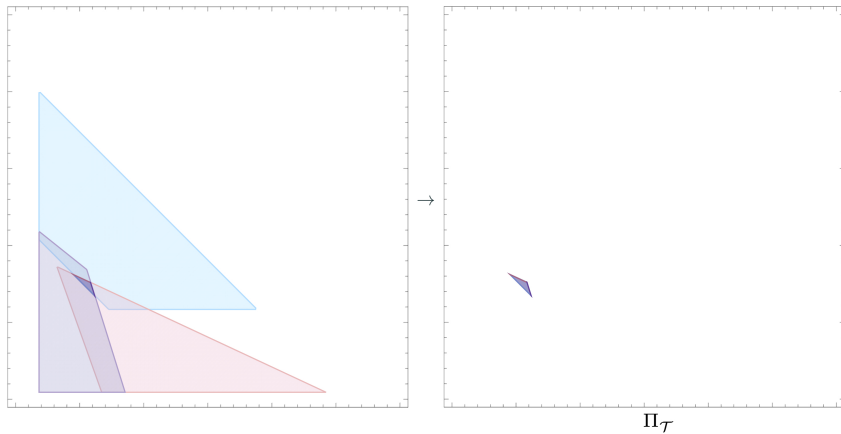
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- Assume potential outcomes exist, impose no further restrictions in Theorem 2 $\implies$ sharp test for random assignment of Z

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- Or, use Fang and Seo (2021) to test (i) and Fang et al. (2023) to test (ii)

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Lemma 2

Suppose $\mathcal{B}_d(\mathbf{p}\mathbf{f}_d(y)) \neq \emptyset, \forall y \in \mathcal{Y}$. Then, for all $(\mathbf{c}_1, \mathbf{c}_2) \in \mathbb{R}^{\mathcal{T}} \times \mathbb{R}^{\mathcal{T}}$,

$$h_{\Theta_{\mathcal{T}|d}}(\mathbf{c}_1, \mathbf{c}_2) = \int_{\mathcal{Y}} h_d(y \mathbf{c}_1 + \mathbf{c}_2; \mathbf{p}\mathbf{f}_d(y)) d\mu(y).$$

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Target Parameter: $\mathbb{E} [Y_d \mid \boldsymbol{T} \in \mathcal{T}_\dagger]$, for some subset $\mathcal{T}_\dagger \subseteq \mathcal{T}$.

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→ for each γ , check if intersection of three **closed convex sets** is empty

→ (under some conditions) can use **hyperplane separation theorem**

Mean Outcomes: Inclusion Test

Theorem 3

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where $\odot$ denotes the coordinate-wise product of two vectors (Hadamard product).

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→ Need to solve one unconstrained convex program per γ

Theorem 4 (Sharp Bounds on T -Conditional Y_d -Means)

Suppose $\mathbb{S}$, d , and $\mathcal{T}_\dagger$ are as in Theorem 3. Define

$$\underline{p}_\dagger := \inf_{\mathbf{x} \in \Pi_{\mathbb{S}}} \mathbf{c}_\dagger^\top \mathbf{x} \quad \text{and} \quad \bar{p}_\dagger := \sup_{\mathbf{x} \in \Pi_{\mathbb{S}}} \mathbf{c}_\dagger^\top \mathbf{x} .$$

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For each $p \in [\underline{p}_\dagger, \bar{p}_\dagger]$, $\mathbf{v} \in \mathbb{R}^{\mathcal{T}(\mathbb{S})}$, and $\lambda \in \mathbb{R}$, define

$$L_d(p; \mathbf{v}, \lambda) := - \int_{\mathcal{Y}} \mathbf{h}_{d(y)}((-y + \lambda) \mathbf{c}_\dagger + \mathbf{v}) \, d\mu(y) - \mathbf{h}_{\Pi_{\mathbb{S}}}(-\mathbf{v}) + \lambda p ,$$

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Suppose $\underline{p}_\dagger > 0$.

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For each $p \in [p_{-\dagger}, \bar{p}_{\dagger}]$, $\mathbf{v} \in \mathbb{R}^{\mathcal{T}(\mathbb{S})}$, and $\lambda \in \mathbb{R}$, define

$$L_d(p; \mathbf{v}, \lambda) := - \int_{\mathcal{Y}} \mathbf{h}_{d(y)}((-y + \lambda) \mathbf{c}_{\dagger} + \mathbf{v}) \, d\mu(y) - \mathbf{h}_{\Pi_{\mathbb{S}}}(-\mathbf{v}) + \lambda p,$$

$$U_d(p; \mathbf{v}, \lambda) := \int_{\mathcal{Y}} \mathbf{h}_{d(y)}((y - \lambda) \mathbf{c}_{\dagger} + \mathbf{v}) \, d\mu(y) + \mathbf{h}_{\Pi_{\mathbb{S}}}(-\mathbf{v}) + \lambda p.$$

Suppose $p_{-\dagger} > 0$. Then, the following bounds are sharp under $\mathbb{S}$:

$$\min_{p_{-\dagger} \leq p \leq \bar{p}_{\dagger}} \frac{L_d(p)}{p} \leq \mathbb{E}_Q[Y_d \mid \mathbf{T} \in \mathcal{T}_{\dagger}] \leq \max_{p_{-\dagger} \leq p \leq \bar{p}_{\dagger}} \frac{U_d(p)}{p},$$

where $L_d(p) := \sup_{\lambda, \mathbf{v}} \{L_d(p; \mathbf{v}, \lambda)\}$ and $U_d(p) := \inf_{\lambda, \mathbf{v}} \{U_d(p; \mathbf{v}, \lambda)\}$.

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$$\int_{\mathcal{Y}} \mathbf{h}_{d(y)}(y \mathbf{c}_{\dagger} + \mathbf{v}_d) \, d\mu(y) + \int_{\mathcal{Y}} \mathbf{h}_{d'(y)}(-y \mathbf{c}_{\dagger} + \mathbf{v}_{d'}) \, d\mu(y) + \mathbf{h}_{\Pi_S}(-\mathbf{v}_d - \mathbf{v}_{d'} - \lambda \mathbf{c}_{\dagger}) + \lambda p.$$

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- If $\mathcal{F}_T(\mathbb{S})$ is a **convex polytope**, there exists $\{\mathbf{c}_{k|d}\}_{k=1}^n$ such that

$$\Pi_{\mathbb{S}} = \left\{ \mathbf{x} \in \mathbb{R} \mid \mathbf{c}_{k|d}^{\top} \mathbf{x} \leq \mathbf{h}_{\Pi_{\mathbb{S}}}(\mathbf{c}_{k|d}) \forall k = 1, \dots, n \right\}.$$

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In such cases, $U_d(p)$ has the following **“multiplier”** representation

$$U_d(p) = \inf_{\substack{\lambda \in \mathbb{R}, \\ \gamma \in \mathbb{R}_{\geq 0}^n}} \left\{ \int_{\mathcal{Y}} \mathbf{h}_{d(y)} \left((\kappa(y) - \lambda) \mathbf{c}_{\dagger} - \sum_{k=1}^n \gamma_{(k)} \mathbf{c}_{k|d} \right) d\mu(y) + \sum_{k=1}^n \gamma_{(k)} \mathbf{h}_{\Pi_{\mathbb{S}}}(\mathbf{c}_{k|d}) + \lambda p \right\}$$

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→ In many cases, $n \ll |\mathcal{T}|$, leading to computational gains

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Simple Bounds: Solving for λ

Differentiating wrt λ and cancelling terms yields the following FOC

$$p = \int_{-\infty}^{\lambda} -h_{d(y)}(-c_{\dagger}) d\mu(y) + \int_{\lambda}^{\infty} h_{d(y)}(c_{\dagger}) d\mu(y) ,$$

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- If Y is not continuous, can use suitable generalized inverse

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→ More generally, $\mathfrak{F}_{\downarrow}^{-1}[f](q) = \mathfrak{F}_{\uparrow}^{-1}[f](\int_{\mathcal{Y}} f(y) dy - q)$

Theorem 5 (Generalized Trimming Bounds)

Suppose $\mathbb{S}$, d , and $\mathcal{T}_{\dagger}$ are as in Theorem 3.

Define $\bar{f}_{\dagger|d}(y) := \mathbf{h}_{d(y)}(\mathbf{c}_{\dagger})$ and $\underline{f}_{\dagger|d}(y) := -\mathbf{h}_{d(y)}(-\mathbf{c}_{\dagger})$,

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Then, for any $d \in \mathcal{D}$,

$$\frac{\tilde{L}_d(p_L^*)}{p_L^*} \leq \mathbb{E} [\kappa(Y_d) \mid \mathbf{T} \in \mathcal{T}_\dagger] \leq \frac{\tilde{U}_d(p_U^*)}{p_U^*},$$

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where

$$\begin{aligned} \tilde{L}_d(p) &:= \int_{\mathcal{Y}} y \underline{f}_{\dagger}(y) d\mu(y) + \int_0^{p - \underline{p}_{\dagger|d}} \mathfrak{F}_\dagger^{-1} [\bar{f}_\dagger - \underline{f}_\dagger](q) d\nu(q), \\ \tilde{U}_d(p) &:= \int_{\mathcal{Y}} y \bar{f}_{\dagger}(y) d\mu(y) + \int_0^{p - \bar{p}_{\dagger|d}} \mathfrak{F}_\dagger^{-1} [\bar{f}_\dagger - \underline{f}_\dagger](q) d\nu(q), \end{aligned}$$

and p_L^* and p_U^* are as follows

Theorem 5 contd.

$$p_L^* := \begin{cases} \bar{p}_\dagger & \text{if } p_L > \bar{p}_\dagger \\ \underline{p}_\dagger & \text{if } p_L < \underline{p}_\dagger \\ p_L & \text{otherwise} \end{cases}, \text{ with } p_L := \inf \left\{ p \in [\underline{p}_{\dagger|d}, \bar{p}_{\dagger|d}] \mid \mathfrak{F}_\dagger^{-1} [\bar{f}_\dagger - \underline{f}_\dagger] (p - \underline{p}_{\dagger|d}) \geq \frac{L(p)}{p} \right\},$$

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In cases where $\underline{f}_\dagger(y) = 0$ for all $y \in \mathcal{Y}$, the bounds simplify to

$$\begin{aligned} \frac{1}{\underline{p}_\dagger} \int_0^{\underline{p}_\dagger} \mathfrak{F}_\dagger^{-1} [\bar{f}_\dagger] (q) \, d\nu(q) \\ \leq \mathbb{E} [Y_d \mid \mathbf{T} \in \mathcal{T}_\dagger] \\ \leq \frac{1}{\underline{p}_\dagger} \int_0^{\underline{p}_\dagger} \mathfrak{F}_\dagger^{-1} [\bar{f}_\dagger] (q) \, d\nu(q) \end{aligned}$$

Comments

- Bounds of Horowitz and Manski (1995), Lee (2009), Kitagawa (2021), and Marx (2024) can be obtained as **special cases of Theorem 5**
- Sharp in many cases, but not in general
 - Sharp if $\begin{bmatrix} \mathbf{R}_d \\ \mathbf{c}_\dagger^\top \end{bmatrix}$ *with duplicate and zero columns removed* is tall
 - In particular, sharp under the **Unordered Monotonicity** condition of Heckman and Pinto (2018)
 - Sharp for Examples 1 & 2, not sharp for Example 3

Remaining Content of Paper

1. **Identification:** Remaining identification results, including under some assumptions on outcomes
2. **Analytic bounds:** Automated derivation of analytic expressions for “point-wise” bounds
3. **Inference:** Support function based test statistic for inclusion tests
4. **Computation:** Primal-dual method for computing support function
 - feasible for large problems, exact solution in finitely many steps
 - can also provide fast approximations with deterministic error bounds

Conclusion

Summary

Framework for partial identification in IV treatment selection models

- multivalued treatments, discrete instruments
- sharp under random assignment of instrument
- tractable, not “black box”: analytic bounds
- sharp specification tests

Current and Future Work

- Formalizing inference and estimation
- Simulations and empirical applications
- Developing a Stata/R/Python package
- Additional guidance on when bounds are likely to be informative

Thank you!

Basic Example ($Z \in \{0, 1\}$, $D \in \{0, 1\}$, no selection restrictions)

For $d = 1$, $(\star)$ implies

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix} \boldsymbol{\xi}_1(y) = \mathbf{p}\mathbf{f}_1(y)$$

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The support function for the 1-dimensional polytope (i.e. interval) defined by the non-negative solutions to the above system is simply the map

$$c \mapsto \begin{cases} c \left(\min_{z \in \{0,1\}} \left\{ P(D=1|Z=z) f_{Y|D=1,Z=z}(y) \right\} \right) & \text{if } c \geq 0 \\ 0 & \text{if } c < 0 \end{cases}$$

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$\Pi_{\mathcal{T}|1}$ consists of all $\pi \geq 0$ such that

$$\pi_{(AT)} \leq \int_{\mathcal{Y}} \min_{z \in \{0,1\}} \left\{ P(D=1|Z=z) f_{Y|D=1,Z=z}(y) \right\} d\mu(y) ,$$

$$\pi_{(AT)} + \pi_{(D)} = P(D=1|Z=0) = \int_{\mathcal{Y}} P(D=1|Z=0) f_{Y|D=1,Z=0}(y) d\mu(y) ,$$

$$\pi_{(AT)} + \pi_{(C)} = P(D=1|Z=1) = \int_{\mathcal{Y}} P(D=1|Z=1) f_{Y|D=1,Z=1}(y) d\mu(y) .$$

Similarly, $\Pi_{\mathcal{T}|0}$ consists of all $\pi \geq 0$ such that

$$\pi_{(NT)} \leq \int_{\mathcal{Y}} \min_{z \in \{0,1\}} \left\{ P(D=0|Z=z) f_{Y|D=0,Z=z}(y) \right\} d\mu(y) ,$$

$$\pi_{(C)} + \pi_{(NT)} = P(D=0|Z=0) ,$$

$$\pi_{(D)} + \pi_{(NT)} = P(D=0|Z=1) .$$

$\Pi_{\mathcal{T}}$ consists of all $\pi \geq 0$ that satisfy the 6 conditions above.

MTO Example: Sharp Bounds on $\pi_{[(\cdot, M)]}$

For the MTO example considered earlier, the following bounds are sharp

$$\max \left\{ \int_{\mathcal{Y}} \max \left\{ 0, \mathbf{p}\mathbf{f}_{M|Z=(1,1)}(y) - \mathbf{p}\mathbf{f}_{M|Z=(0,0)}(y) \right\} d\mu(y), \right. \\ \left. \mathbf{p}_H[(0,1)] - \mathbf{p}_H[(1,1)] \int_{\mathcal{Y}} \max \left\{ 0, \mathbf{p}\mathbf{f}_{L|Z=(0,0)}(y) - \mathbf{p}\mathbf{f}_{L|Z=(1,1)}(y) \right\} d\mu(y) \right\}$$

$$\leq \pi_{(HHM)} + \pi_{(HLM)} + \pi_{(LLM)} \leq$$

$$\min \left\{ \mathbf{p}_M[(1,1)] - \mathbf{p}_M[(0,1)], \right. \\ \left. \mathbf{p}_M[(1,1)] - \mathbf{p}_M[(0,0)] + \int_{\mathcal{Y}} \min \left\{ \mathbf{p}\mathbf{f}_{L|Z=(1,1)}(y), \mathbf{p}\mathbf{f}_{L|Z=(0,1)}(y) - \mathbf{p}\mathbf{f}_{L|Z=(0,0)}(y) \right\} d\mu(y) \right\}$$

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